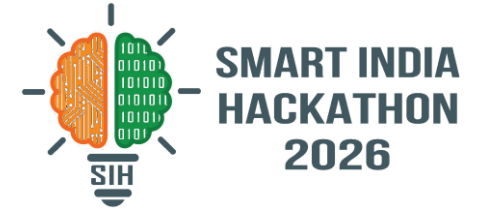


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID - SIH26165
- Problem Statement Title - AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act / Unsafe-Condition and Near-Miss Reports
- Theme - Smart Automation
- PS Category - Software
- Team ID - _____
- Team Name (Registered on portal) - Signal-0

Working prototype: <https://sanket-frontend.onrender.com>

IDEA TITLE

SANKET — finding the reports that could have killed someone

Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- A worker types or speaks a safety report. The AI reads it and asks three questions.
- Is there a serious hazard? Was a safety control missing or broken? How bad could it have been?
- If all three point the wrong way, the report is flagged and moves to the top of the queue.

How it addresses the problem

- OIL gets more reports than anyone can read carefully, so the dangerous ones get lost in the pile.
- SANKET reads every one and reorders the pile. It never closes a report — a person still decides.

Innovation and uniqueness of the solution

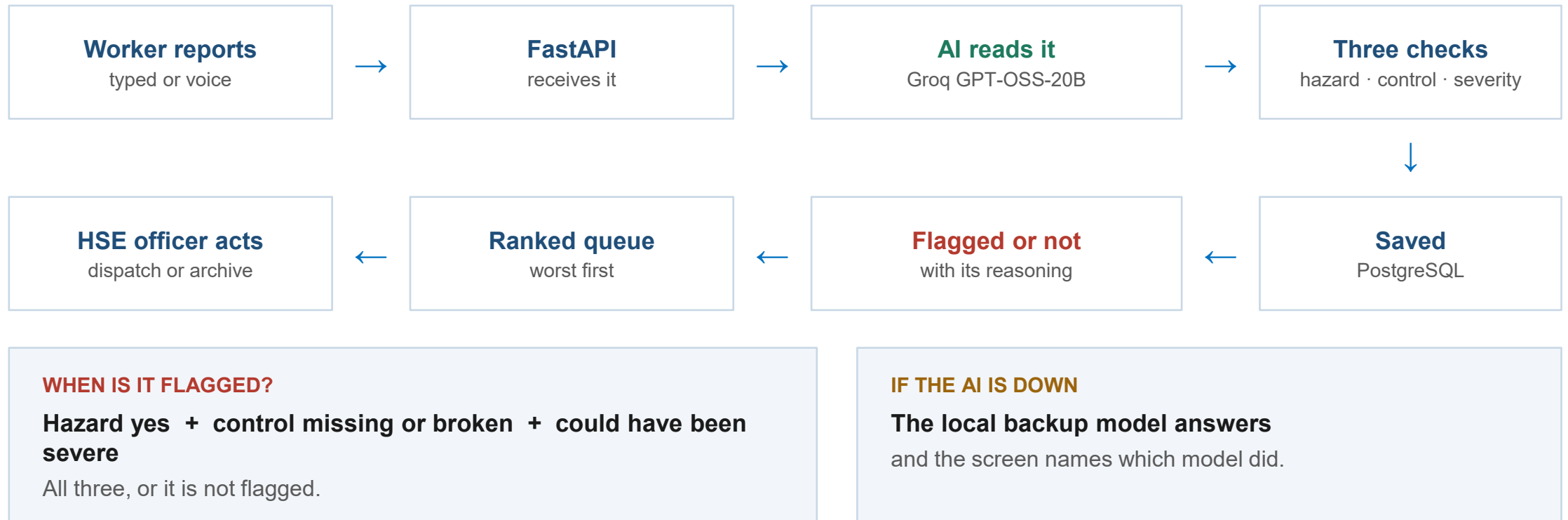
- Three separate checks instead of one risk score, so you can see which one drove the answer.
- If the AI is unreachable a small local model answers, and the screen says which one answered.

TECHNICAL APPROACH

Technologies to be used (e.g. programming languages, frameworks, hardware)

- Frontend: React Backend: FastAPI (Python) Database: Supabase (PostgreSQL)
- AI model: Groq GPT-OSS-20B Backup model: TF-IDF + Logistic Regression, runs locally

Methodology and process for implementation (Flow Chart)



FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

- Uses the safety reports OIL already writes — no new data collection needed.
- No special hardware. Runs on an ordinary web server.
- A working prototype is already built and live.

Potential challenges and risks

- Every person writes a report differently.
- Many reports never say whether a safety control was in place.
- The AI service can be slow, busy or unavailable.
- An AI answer nobody can check is not worth trusting.

Strategies for overcoming these challenges

- A written rulebook that both the AI and our human reviewers follow.
- "Not stated" is a valid answer for the control question — we never guess.
- A local backup model, and every answer records which model produced it.

IMPACT AND BENEFITS

Potential impact on the target audience

- HSE officers read the dangerous reports first, not in the order they arrived.
- Managers see which sites and shifts keep producing near-misses.
- Workers get an answer in seconds, so reporting feels worth doing.

Benefits of the solution (social, economic, environmental, etc.)

- Social — a warning acted on before someone is hurt or killed.
- Economic — less time screening reports by hand, fewer incidents, less downtime.
- Environmental — the same failures that injure people also cause spills and leaks.
- Organisational — every judgement is recorded, so audits are straightforward.

Safety Reports → Precursors Found → Priority → Early Action → Safer Operations

Details / Links of the reference and research work

- IOGP Life-Saving Rules — the hazard categories we tag against
- DEKRA SIF Framework — serious injuries have identifiable precursors
- EEI SIF Model — a precursor is a hazard with a missing or failed control
- OSHA Severe Injury Reports — 30 real published narratives, used to test the system on writing nobody on our team wrote

How we tested it

- 150 safety reports written for this prototype, each reviewed by two people using the same rulebook.
- On a 38-report sample the two reviewers worked separately and agreed 97% of the time (Cohen's kappa 0.947).
- No system should claim to be more consistent than the people who made its labels.
- Our simple backup model scores F1 0.75. The AI model is being re-measured after we found and fixed a flaw in our own test setup.

Live prototype: <https://sanket-frontend.onrender.com>
GitHub: SANKET — AI-powered SIF precursor detection